Module 8
Illness and Injuries

Injuries and illnesses happen, and as a coach, you need to know how to deal with these situations as they arise and know how the training process may be impacted. Running is a sport with a relatively high injury rate, and as such, understanding methodologies to possibly prevent injuries is of critical importance.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Illness/Injury terminology
- What constitutes an emergency situation
- Aspects of injury prevention
- Running injuries
- Tendonitis or tendinosis?
- Role of running surface on impact forces
- Effect of colds/flu on an individual
- Heat-related illnesses
- Hypothermia
- Altitude sickness

❖ INJURY

TERMINOLOGY

Sprain: Injury to a ligament (ligaments connect bone to bone). Sprains are categorized as Level 1–3, 3 being the most severe.

Strain: Injury to muscle or tendon (tendon connects bone to muscle).

Muscle Cramp (exercise-based): Involuntary spasm of a muscle or muscles. Cramps can be caused by many factors, including overuse of a muscle in duration and/or intensity and nerve-related issues. Mild dehydration and low electrolyte levels, which were previously associated with cramping, have largely been determined not to be a valid catalyst of cramps (763–765).

Fracture: A break in a bone; varying degrees of severity.

Tendinosis: (often incorrectly called "tendonitis") Damage to a tendon. Often caused by overuse.

Bursitis: Inflammation of a bursa sac. A bursa sac is a synovial fluid-filled sac that is located between bone, muscle, tendons, etc. The purpose of a bursa sac is to provide cushioning and reduce friction.

Nerve Impingement: A nerve that has pressure applied to it. Often referred to as a *pinched nerve* or *compressed nerve*.

RICE: Acronym for ***R****est*–***I****ce*–***C****ompression*–***E****levation*.

Chafing: Location on the body where the top layer of skin is irritated or worn away either due to skin-on-skin rubbing (e.g., thighs) or more commonly from skin rubbing against clothing (e.g., bra straps). While not an injury exactly, it can be very painful and, like an open wound, it can get infected if not treated. The most common places for chafing are the nipples (men), inner thighs, areas where the bra rubs (women), and under the arms.

To reduce the chance of chafing, here are some preventative measures:

- Try new clothes on short runs before longer runs
- Men: wear nipple covers
- Women: make sure the bra fits properly and is tight
- Apply a skin lubricant such as *Body Glide*® in areas that are likely to get chafed
- Clothing with hidden seams are preferable to garments with regular seams

Your clients will experience discomfort and pain from time to time. It is the responsibility of both you and your client to assess the situation, use the best judgment, and determine whether medical intervention is necessary or if the issue is something more benign. If you or your client is unsure if medical intervention is necessary, seek medical intervention! What follows is a list of possible scenarios and the advised courses of actions.

INJURY SCENARIOS

Discomfort that appears to be due to a biomechanical issue:
Work to correct the biomechanical issue to see if the discomfort diminishes or disappears. If not, refer to a specialist. If a biomechanical issue is substantial or if you do not feel comfortable addressing the issue, refer immediately to a specialist.

Acute pain or trauma that is severe:
Refer to a specialist or emergency care personnel.

Pain that appears to be the result of overuse:
Reduce duration and intensity. If pain does not diminish or disappear, refer to a specialist.

Random pain:
Refer to a specialist.

Discomfort or pain due to illness:
Cease physical activity, and depending on the severity of the illness, the client should contact a doctor.

Chronic Pain:

Chronic pain is characterized as pain that has been present for six months or more. It is advised to have the client see a medical professional.

You must work within your scope of knowledge and practice at all times. In other words, know and respect your limits. If at any time you feel that your client's situation is at the threshold of your knowledge or practice, you must refer the individual to a specialist. As such, this certification will not cover how to diagnose or treat injuries, as this constitutes an area that is outside the scope of practice of a running coach.

While diagnosis and treatment of injuries are outside the scope of a running coach, it does not mean that you should be unfamiliar with the most common types of injuries and their potential symptoms.

The only advice that you can give a client in regard to treatment is **R**est–**I**ce–**C**ompression–**E**levation (RICE). Beyond RICE, recommending treatments as seemingly benign as heat need to be avoided. Aside from the fact that it is outside the scope of your practice, it is also important to note that just because a "treatment" has worked for you, it may not necessarily work for someone else. Additionally, you cannot recommend or prescribe medications. This includes over-the-counter medications such as ibuprofen and other anti-inflammatories.

Emergency Situations

If any of the following situations arise, emergency personnel must be alerted immediately.

Note: Contacting emergency personnel is not limited to the following situations. Do not leave the individual and/or scene until emergency personnel arrive.

- Broken bone
- Loss of consciousness
- Shock
- Choking
- Cardiac arrest
- Drowning
- Respiratory event (e.g., asthma attack)
- Allergic reaction

- Unresponsiveness
- Uncontrolled bleeding

While coaching clients in person, you should always have a cell phone available to call emergency personnel (#911).

Asthma: Inhaler Use

If your client is asthmatic and, in particular, suffers from exercise-induced asthma attacks, it is advised the person carries a fast-acting inhaler while exercising.

Contents of an inhaler are a prescription medicine. Therefore, you are not legally allowed to administer the medicine for clients – even if they are suffering an asthma attack. Clients **MUST** administer the drug/inhaler themselves.

INJURY PREVALENCE AND LOCATION

A study by Van Mechelen found that running injuries among recreation runners ranged from 37–56 percent annually (192). Other pertinent findings from this study:

- Most running injuries involve the lower extremities
- 50–75 percent of injuries are due to overuse (repetitive stress)
- Recurrences of running injuries range from 20–70 percent

The Van Mechelen study also found that many of the commonly associated risk factors for running injury are unclear, in large part due to contradictory or scarce research findings. Some of these factors include:

- Warm-up/Stretching
- Muscular imbalances
- Lack of range of motion
- Orthotics

Therefore more research is required to assess the validity of these areas regarding their correlation to running injury.

Many factors influence injury prevalence and location. A 2012 study by Daoud et al. examined repetitive stress injuries among collegiate runners and found that 74 percent of runners experienced a moderate to severe injury annually (648). Some of the study's specific findings:

- The most common injuries were:
    - Shin splints
    - IT band syndrome
    - Patellofemoral syndrome
    - Achilles tendon pain

Runners in this study ran 40–45 miles/week, performed high-intensity training sessions, and ran between 12 and 18 races per year. As such, runners in this study are classified as competitive versus recreational, and therefore it is probable that factors such as high mileage, fast running speeds, and frequent racing contribute to an increased injury rate (648).

Influence of Foot-Strike Type

The aforementioned study (Daoud et al.) found that in regard to repetitive stress injuries, runners who ran with a rear foot strike were two times more likely to sustain a repetitive stress injury than those who ran with a midfoot strike (648).

Anatomical Factors

A 2010 study examining factors involved in preventing injuries found that two anatomical factors correlate highly with running injuries (649):

1. High-arched feet
2. Leg-length inequality (functional or structural)

‘Hard on the Knees’

Running is a sport that is often associated with being *hard on the knees*. The common association is that compressive forces on the knees due to the impact nature of running cause knee injuries.

A 2013 study by Hinterwimmer et al. examined the effect of marathon training in regard to cartilage changes (volume and thickness) of the knees of beginner marathoners (690). It was found that while there were some cartilage changes, they were not clinically significant and therefore long-distance running is well tolerated by the cartilage in the knees of beginner marathoners. More research is required regarding changes/loss of cartilage in seasoned marathon runners.

Dr. Michael Fredericson of the Stanford Running Clinic notes (772),

“At faster speeds, runners tend to have better hip mechanics, which leads to reduced loads on the knee.”

This primarily has to do with hip rotation in the transverse plane.

A recent study by Petersen et al. also found that running at slower versus faster speeds increased the load on the knees. This, however, was due to a decrease in the overall number of strides to cover the same distance (773).

Lastly, Dr. Fredericson states that running with a foot strike closer to the front of the foot reduces the cumulative load on the knees.

Ultrarunners

Ultrarunning is defined as running any distance beyond that of a marathon (26.2 miles). As high mileage substantially increases one’s chance of sustaining an injury, it should not come as a surprise that ultrarunners have a relatively high rate of injury.

A study of 1,212 ultrarunners returned the following results (650):

- Three-quarters of ultrarunners reported having exercise-related injuries the prior year. Of these, 65 percent lost at least one day of training because of injury.
- Injured ultrarunners tended to be younger and less experienced.
- 3.7 percent reported having a stress fracture.
  - 48 percent of stress fractures are in the feet.
  - Stress fractures among ultrarunners are less than in non-ultrarunners.

INJURY PREVENTION

While injuries occur for various reasons, you can take measures to mitigate this risk for your clients.

Volume Increases

Traditionally, a 10 percent maximum increase in weekly training volume and a 10 percent maximum increase of the longest training day (week-to-week) are recommended. The thought process behind the *10 percent rule* is that volume increases greater than 10 percent will increase the chance for injury.

The viewpoint of this certification program is that while 10 percent is ideal for some clients, it may be too high or too low for others. Therefore, while the 10 percent rule is a general guideline, changes in training volume must be based on feedback from a client. If a client is starting out at the absolute lowest training volume, initial volume increases can typically afford to be larger than 10 percent.

A study by Fields et al. found that regarding preventative injury strategies, only reducing weekly running mileage correlated strongly with decreased running injuries (649).

Intensity

The level of intensity and the number of intensity-based workouts should increase incrementally. The more intense a workout is, the greater the demand on the musculoskeletal system. Without proper progression, muscles and connective tissue will not have the proper amount of time to adapt to the increased workload, thereby increasing the risk of injury. When intensity is initially added to a program (e.g., hill repeats, fartleks, repeats, etc.), only one day should be allocated to this purpose. As the body adapts, additional intensity days can be added.

Cross-training

By definition, cross-training is when individuals participate in a sport or activity other than the sport they compete in for the purpose of performance enhancement. Common cross-training activities are:

- Resistance training
- Cycling
- Swimming/water jogging
- Yoga and Pilates
- Other cardiovascular (e.g., rowing, elliptical, boxing, soccer, basketball, cross-country skiing)

While these are some of the more popular activities, any activity other than running is considered cross-training. As most of the movements involved with running occur in the sagittal and transverse planes, cross-training in the frontal plane is recommended.

A 2013 study by Malisoux et al. found that cross-training is a viable strategy to reduce the chance of running injury due to varying the load applied to the musculoskeletal system (687).

Cross-training has many benefits, including increasing cardiovascular fitness, injury prevention, increasing strength, staying mentally fresh, and, perhaps most important— having fun!

Fatigue

When individuals become fatigued, their form can break down and may lead to injury. During training and racing, a client must always be conscious of fatigue level. If a client’s fatigue gets to the point where form breaks down significantly or the person cannot maintain focus and clarity, the activity level needs to be reduced or ceased. In addition, it is likely a client needs to refuel when the energy level drops substantially. If the fatigue level is so high that an individual's safety is at risk, exercising must cease.

Kinesio Tape

Kinesiology tape (often called *Kinesio Tape*) is a stretchable adhesive tape that is primarily used for injury prevention/intervention or to increase performance. This tape was developed by Dr. Kenzo Kase in 1973 (214). Specifically, kinesiology tape is used for muscle and joint support, enhancement of biomechanical efficiency, and to aid in muscle recovery and endurance.

Studies researching the benefits of kinesiology tape have resulted in largely inconsistent results (207, 209). It is clear that more research is needed in this area to determine the effectiveness of the tape in regard to its purported benefits. It can be assumed that use of kinesiology tape will not diminish performance or reduce joint stability or functioning, and it could possibly enhance these areas (208). Therefore, if clients ask about kinesiology taping or want to try it, it is best to refer them to a specialist who has experience in this area.

Ice

It has long been advised to ice not only injured areas, but also to ice areas on the body to prevent injury. The genesis behind this was primarily to reduce inflammation. However, it is now believed that inflammation is a critical aspect of the healing and muscle-repair process.

A 2015 study by Leeder et al. found that an ice bath did not promote recovery in subjects with DOMS (678). In fact, icing may slow the healing process. Dr. Jonathan Leeder stated that there is solid evidence to show that athletes feel better after an ice bath and this is likely why ice baths are popular and are believed to correlate with muscle healing and recovery. Despite this, there is no evidence to show that icing results in increased performance and enhanced recovery (678).

In regard to injury, immediate icing is likely beneficial. However, the advice of a medical professional should always be sought and followed.

Running Shoes

While there is no evidence-based research to demonstrate that running shoe type reduces injury, there is research that suggests that swapping out different types of running shoes during training reduces the risk for injury (687). It is theorized that by using different shoes types throughout the training process, the potential for injury is reduced because of the variation of load applied to the musculoskeletal system.

RUNNING INJURIES

Running is synonymous with injury because of one primary factor – impact. According to Van Gent et al., between 37 and 56 percent of runners sustain at least one injury annually and between 20 and 70 percent of those seek medical attention (191). Another study found that between 30 and 90 percent of running injuries caused runners to decrease or stop their training (192).

Two studies, Marti et al. (1987) and Taunton et al. (2002), examined runners to identify risk factors that lead to running injuries. The most common injuries involved the knees and feet. Interestingly, both studies came across four factors that placed runners at greater risk for injury (205, 206):

1. Running high mileage (25–37 miles/week)
    - Mileage greater than this did not increase the chance for injury.
2. Competitive runners with a dedicated focus to training and racing
3. Younger than 34
    - Runners over the age of 34 were at greater risk for Achilles and calf injuries.
4. Newly active (active less than 8.5 years)

Other Factors (205):

- **Weight:** This follows a bell-shaped curve, meaning unless runners were substantially under or overweight, they were not at an increased risk for injury.
- **Shoes:** Shoe type did not affect injury rate.
- **Gender:** Women are at a higher risk of injury than men.
    - **Highlighted Gender-Specific Injury Areas:**

- **Women:** knee, hip, shin
- **Men:** patellar and Achilles tendons

These risk factors correlate to a 2014 study by Kluitenberg et al. that examined risk factors for novice runners. This study had 1,696 participants, ranging in age from 18–65. Of the novice runners, 10.9 percent sustained a running-related injury and the identified risk factors based on those injured were as follows (647):

- Older runners
- Higher *body mass index (BMI)*
- Previous musculo-skeletal complaints not related to running
- No prior running experience

What Is Tendonitis?

Among endurance athletes and especially runners, the term and diagnosis of **tendonitis** is quite common. To understand what this term means, attention needs to be paid to the suffix *-itis.* This is a pathological/medical term that denotes inflammation. Therefore *tendonitis* means inflammation of a tendon.

The issue with the term *tendonitis* is that recent research has shown that in conditions diagnosed as tendonitis, there is little to no inflammatory cells present (606). Therefore the term **tendinosis** is more applicable. The primary finding regarding the cause of tendinosis is a disruption of collagen fibers of tendon, not inflammation. Because of the disruption of collagen fibers, the tendon weakens and is unable to support high loads (603, 604). Collagen is a structural protein in connective tissue, such as tendons.

As there is likely no inflammation present with tendinosis, common treatment methods such as anti-inflammatory medications and corticosteroid injections have been found to increase recovery time and accelerate the degenerative process (603-606). Eccentric strengthening has been shown to assist in increasing the collagen and strength of a tendon (603, 604).

If a client believes he or she has tendinosis, it is advised to refer the person to a specialist.

Common Causes of Running Injuries

Common factors can cause or exacerbate an injury due to stress. In addition to and including some of the aforementioned risk factors, following are the five potential risk factors in regard to running injury.

1. **Poor Running Mechanics**
   While some mechanical issues can be easily identified, such as foot-strike type and excessive pronation or supination, it does not mean they are easy to change. This is because poor running mechanics are often the result of muscle imbalances and kinetic chain issues. Retraining the body to run "properly" is not an overnight solution but rather a long process.

2. **Worn Shoes**
   This is primarily an issue for those who heel strike, as a midfoot strike uses primarily knee flexion for shock absorption rather than heel padding. As noted previously, while the body adapts for reduction in shoe cushioning, excessive wear is likely a contributing factor for injury – especially when an individual utilizes shoes for biomechanical foot and ankle support.

3. **Lack of Range of Motion**
   Lack of range of motion, whether it be the hamstrings, lower back, quadriceps, or any other muscle group, will negatively affect an individual. Tight hamstrings are often the primary cause of injury in regard to a lack of flexibility. This is especially true with those who run with a heel strike, as this stresses the hamstrings eccentrically.

4. **Mileage and Intensity**
   Starting a training program with mileage greater than what a client is ready for, or increasing mileage too fast, is a common cause of injury. Additionally, integrating speed work into a training program before a client is ready for it can also cause injury. Lastly, doing too many days of speed work or high mileage during a week can result in excessive stress on the musculoskeletal system. This often results in a runner not being able to recover sufficiently and places the individual at higher risk for injury.

   Metabolic fitness precedes structural readiness (190). This essentially means that one's aerobic capacity/fitness develops at a faster rate than muscular and connective tissue strength. Therefore, runners tend to increase before their bodies can handle it.

Like incremental increases in distance, intensity must progress at the correct rate as well. Intensity should be introduced into a training program at a relatively low starting point.

5. **Muscle Imbalances**

   This goes hand in hand with *poor running mechanics*. Minor muscle imbalances are normal, as hardly any synergistic muscle groups are equal in strength and flexibility. However, substantial muscle imbalances are not only inefficient but are also potentially injurious to a runner.

Running injuries can range in severity from a minor muscle strain to a major issue, like a fracture. The key to running injury-free is to respect the impact on the body and therefore perform due diligence in regard to the training process. Following are areas to take note of when prescribing a running program:

- Always warm up properly.
- Be aware of high-volume increases.
- If performing speed work, start with only one day per week and no more than two days per week, maximum.
- Work to have your client run with a proper running gait.
- Do not start speed work until a solid base mileage has been established.
- If your client experiences muscular or skeletal discomfort on a run that is not a normal byproduct of the intensity of the workout, have the person stop and RICE, and if the pain continues, refer the individual to a specialist.
- Allow for enough rest in the training program.
- If your client is injured and has sought a doctor’s advice, make sure the person follows the advice, even if it seems that the individual is “healed” sooner than the doctor thought.
- Cross-train.
- Train your client in frontal and transverse planes, not just the sagittal plane.
- Listen to the body (e.g., if clients sense they are still very sore from a previous workout, it is advisable to rest or cross-train versus run – even if it is not scheduled in the program).

Even if clients follow all the “rules” of safe running, there is no guarantee they will not get injured. However, training intelligently mitigates the risk of getting injured, and at the end of the day, this is the best one can hope for.

Common Running Injuries

- **Iliotibial Band Syndrome (commonly referred to as *ITB syndrome*)**
    - This causes pain on the lateral aspect of the knee. The primary cause is the IT band rubbing against the lateral femoral epicondyle (bony protrusion at the base of the femur) repeatedly to the extent that inflammation and pain occurs.

- **Runner’s Knee (commonly called *patellofemoral syndrome,* or *PFS*)**
    - This is a generic term used to describe pain under the kneecap. This is caused by tight connective tissue as well as patellar malalignment. Patellar malalignment is often found in clients with a large Q angle (89). Compressive and shearing forces as well as hip rotation asymmetry have been shown to potentially cause PFS (343). Tight quadriceps and IT bands as well as a lack of gluteal strength contribute to patellofemoral syndrome.

- **Shin Splints**
    - Shin splint is also a relatively generic term used to describe pain in the region of the tibia. Shin splints can be categorized into three main areas: muscular, tibial fascia, and skeletal. If clients experience shin pain, it is advised to have them seek out medical advice, as the type of treatment will likely differ based on the “type” of shin splint.

        - **Muscular:** This is often referred to as *exertional compartment syndrome.* In the case of shin splints, the tibialis anterior is the muscle affected. During exercise, increased blood flow increases the size of the tibialis anterior. However, like all muscles, the tibialis anterior is encased in fascia and can increase in size only so much. This increase in pressure on the fascia is what causes pain. Individuals who have tight fascia surrounding the tibialis anterior are the ones who most often experience exertional compartment syndrome (385). Typically once one stops running, the pain gradually subsides.

As the tibialis anterior dorsiflexes the foot/ankle, when an individual experiences exertional compartment syndrome, the ankle likely has less ability to plantar flex. This lack of plantar flexion results in decreased shock absorption. It could be presumed that this decrease in foot plantar flexion would put more stress on the tibia, thereby increasing the chance for injury to the tibia.

- **Skeletal:** This relates to injury or stress to the bone (tibia). This is commonly referred to as *medial tibial stress syndrome* (MTSS). However, this term is also used to denote connective tissue/fascia related to shin pain. This typically presents with pain on the inside of the tibia bone.

  If the pain is on the front of the tibia (tibial spine), the individual should seek out a physician, as this type of injury is typically classified as more serious (420).

  - Continuing to run with a shin splint can lead to a stress fracture (small crack in a bone).

- **Tibial Fascia (Tibial Fasciitis):** Inflammation of the deep fascia surrounding the tibia can result in shin pain (442). This is often confused with bone pain.

- **Hamstring Injuries**
  - Hamstring injuries are often caused by hyperextension of the knee. The hamstrings act to eccentrically decelerate the lower leg prior to the support phase of the gait and to extend the hip during the drive phase.

  - Runners who heel strike versus midfoot strike are at a higher risk to have hamstring issues. This is because with a heel strike, the stride of the runner is typically longer and therefore the lower leg moves through a larger range of motion than with a midfoot strike. This large range of motion requires a substantial eccentric contraction on the part of the hamstrings to decelerate the lower leg.

- As the hamstrings contract eccentrically during the swing phase of the running gait, if the pelvis is tilted anteriorly when running, it will put more tension on the hamstrings. Therefore running with a substantial anterior pelvic tilt will likely increase the chance for hamstring injury because of the increased tension on the hamstrings during the swing phase of the gait cycle.

- Running too fast or with too much force puts substantial concentric strain on the hamstrings and can result in injury (hamstring strain).

- **Ankle injuries**
  - Runners who “twist” their ankle are often susceptible to twisting it again. A twisted ankle is also commonly referred to as a *rolled* or *sprained* ankle. This involves spraining of one or more of the ligaments in the ankle. Research done by Gehring et al. demonstrated that the cause of lateral (inversion) ankle sprains is not just isolated to the ankle (310). Compromised movement patterns of the pelvis and femur as well as neuromuscular reflexes contribute to lateral ankle sprains (310).

  - Achilles tendinosis refers to damage of the Achilles tendon.

- **Popliteus Injury**
  - The *popliteus muscle* is a small and relatively overlooked muscle behind the knee that has a profound effect on running.

  - The popliteus has three primary roles:
    - To “unlock” the knee (flex) from a straightened position
    - Prevent knee hyperextension
    - Laterally rotate the femur when the foot is planted

  - Injury to the popliteus can occur as a result of overpronation of the foot/ankle complex and knee hyperextension. When the popliteus is injured it shortens, and as a result, full knee extension is often not possible or only with pain.

- **Plantar Fasciitis:**
  - Plantar fascia is a thick band of fascia that begins at the back of the heel and extends under the foot to the heads of the metatarsals.

Pain associated with plantar fasciitis is most commonly found at the beginning of the arch of the foot, closest to the heel. When running, pain most commonly occurs when the foot pushes off the ground. Tight calf muscles often contribute to plantar fasciitis. Additionally, weak intrinsic foot muscles may increase the load on the plantar fascia (600).

If you suspect clients have any injury, including those just noted, you must advise them to seek out a sports medicine specialist.

What Is a Side Stitch?

While not an injury exactly, a side stitch can be quite painful and can result in the cessation of exercise. Also known as *exercise-related transient abdominal pain* (ETAP) (437), ETAP is most commonly associated with running (437).

ETAP is characterized by a cramping or sharp pain toward the bottom of the rib cage and most commonly occurs on the right side of the body.

While there are many theories as to why ETAP occurs, no one cause has been identified (437). Some of these theories are a lack of electrolytes (438), stretching of ligaments associated with the diaphragm (437), and shallow breathing (438). Another study noted that individuals who demonstrated back kyphosis (excessive flexion) when running were more apt to experience ETAP versus those without kyphosis (439). The most common treatments are stopping the exercise, stretching, and deep breathing.

Prevention of side stitches often involves strengthening the inner-core musculature, slowly easing into increasing running mileage and intensity, increasing one’s fitness level, and getting a proper warm-up.

It should be noted that some individuals experience pain in the shoulder (most often the right shoulder) when running, which is thought to be referred pain from the diaphragm via the phrenic nerve (437).

Other Running-Related Injuries

- **Bunion (hallux valgus)**

  A bunion is an abnormal bone growth that causes the big toe to slant inward toward the other toes. This slant of the toe causes the outside of the big toe joint (first metatarsophalangeal joint) to jut outward (medially). A bunion causes pain and in severe cases requires surgery (651).

  Some evidence shows that wearing tight shoes (in the toe box area) may contribute to bunions. As a result, it is advised to run in shoes that have ample room in the toe box (651).

- **Morton's Neuroma**

  Commonly associated with pain in the middle of the ball of the foot, Morton's neuroma is caused by thickening of the tissue around a nerve in the ball of the foot (652). The pain is often in the form of a burning sensation.

  Morton's neuroma has been associated with wearing high-heeled shoes. As a result, wearing a shoe with a lower heel and wearing a shoe with a wider toe box may help (652).

- **Sesamoiditis**

  Under the big-toe joint exists two pea-sized bones (sesamoids). The sesamoids are embedded in a tendon to assist with the mechanical efficiency of the foot when running or walking (653).

  The pain associated with *sesamoiditis* is due to irritation of the tendon in which the sesamoid bones are embedded. Rest as well as placing a pad on the fore/midfoot with a cutout for the big-toe joint (termed dancer's pad) can assist in taking pressure off the sesamoids and thus enhancing recovery (653).

Running Recovery

If your client has not been running prior to the inception of the training program, you will need to be sensitive to the recovery aspect of running because of the physical stress placed on the body. As such, during a recovery period, running should not be done on back-to-back days, and if your client has a strain or sprain, there must be complete recovery before resuming running. Therefore, if there is any lingering pain, your client should hold off on resuming running. If your client's muscles are *tight*, sometimes a low-intensity run is fine. But you need to understand what your clients mean if they say their legs are "tight." If you are unsure, err on the side of caution and pass on the run.

If your client has access to a pool, water jogging is a common cross-training practice that stimulates the same muscles in more or less the same range of motion as running (outdoors), although with substantially reduced impact on the body. Additionally, a specialized treadmill (*Alter G*®) allows individuals to run at varying percentages of their body weight. Cycling out of the saddle also positions the body similarly to running.

If your client is new to running, incorporating run/walk days into the program is often beneficial. When returning from an injury, slow and proper progression is of the utmost importance. For example, your client might run for one minute and then walk for two minutes, then repeat this cycle four more times. The primary purpose of run/walk programs is not for cardiovascular conditioning, but rather adapting the body slowly to the musculoskeletal demands of running.

As previously noted, nonsteroidal anti-inflammatory drugs (NSAIDs) such as *ibuprofen* and *aspirin* have been found to slow muscle recovery and muscle growth (134).

A 2015 study by Da Silva et al. looked at the effect of taking *ibuprofen* before running on male runners who had preexisting exercise-induced muscle damage (676). The study found that taking ibuprofen did not help prolong the subject's run time. Therefore it is not suggested that runners take NSAIDs prior to running, even if they have sore muscles.

Additionally, a 2013 study by Kuster et al. found that ingesting over-the-counter pain relievers prior to a marathon resulted in increased gastrointestinal issues and muscle cramps (677). The higher the dose of painkillers, the greater the incidence of the aforementioned issues. Some of the participants who took painkillers prerace experienced kidney failure (3), bleeding (4), and cardiac infarctions (2). None of the subjects who did not take painkillers prerace were admitted to the hospital. Based on the aforementioned studies, NSAIDs are not recommended for use prior to competition unless directed by an individual's physician.

Running Surface

Best	Grass
	Dirt
	Rubber Track
	Asphalt
Worst	Cement

Running surfaces are often discussed from the perspective of injury prevention and intervention. A common theory on running surfaces is that there is a hierarchy from worst to best in relation to their respective impact on the body.

Based on this hierarchy, running coaches often advise running on grass or dirt to prevent injury and for those coming back from injury. Several studies examined the plantar load on the feet in relation to running on overground surfaces (asphalt, concrete, grass, rubber track) (278, 279, 280). The findings were largely consistent that grass produced the least amount of pressure on the feet, while foot pressure was similar on the other surfaces. Rubber running track surfaces lose some of their shock-absorption qualities as they age. Therefore, the age of a track will likely be a factor in the pressure on the feet.

Another study looked at the plantar load on the feet when running on the following surfaces: treadmill, concrete, and grass (281). Of the three surfaces, the treadmill resulted in the lowest magnitude of plantar pressure. Therefore, treadmill running might be a good option for runners coming back from injury as well as those looking to run with less impact. Treadmill brands as well as the usage of a treadmill will likely influence the degree of shock absorption. As noted previously, a special type of treadmill, *Alter G®*, allows users to run at a percentage of their body weight, thus lessening the impact on the body.

While grass surfaces seem to be the best to run on for shock absorption, this surface is often uneven and therefore it can be theorized that there is also an increased chance for "rolling" an ankle. Conversely, running in a controlled manner on uneven surfaces may strengthen connective tissue in the feet and ankles because of an increased stability requirement.

Blisters

Blisters are a common running injury. A blister is characterized by a bubble of skin with fluid on the inside. There are two common causes of blisters:

1. Friction between the skin and sock or the skin and the shoe
2. Wet feet due to water or excessive sweating

Blisters may or may not be painful. If your client is running in a race and is experiencing a painful blister, have the person seek out attention at a medical tent.

Ways to prevent blisters (292):

- Make sure feet are dry
- Wear socks that wick sweat
- Do not shave off calluses, as these act to protect the feet
- Run only in properly fitted shoes
- Do not lace up shoes too tight
- Use a small amount of skin lubricant such as *Vaseline*® or *Body Glide*® in areas that are prone to blistering.

Summary

While more research needs to be done in the areas pertaining to the causes and prevention of running injuries, it is important to be aware of factors that may contribute to them. It is important to keep in mind that clients must be assessed on an individual basis. For example, your client may present with many of the proposed running injury risk factors but never gets injured, whereas another client may seemingly be perfect in regard to mechanics and has no significant risk for running injury – but experiences a high rate of injury.

While many studies contradict each other regarding potential causes for injury, the two factors that most all studies agree on as significant contributing factors for running injury are high mileage and increasing mileage too fast.

❖ ILLNESS

Dealing with illness is unfortunately part of the training process. This section will focus on guidelines as well as common environmental factors that can cause illness.

TERMINOLOGY

Hyponatremia: Electrolyte imbalance (i.e., sodium) where the levels of sodium are lower than normal. This can be a result of overhydration.

Dehydration: Occurs when the body takes in less fluid than it uses.

Hypothermia: Occurs when the body's core temperature drops below what is necessary for body function and metabolism to take place at a normal rate.

Heat Illnesses: A range of heat-related illnesses that can range from fairly benign (e.g., cramps) to heat stroke, which can be fatal.

COLDS/FLU

This is a common ailment for many distance runners. Endurance exercise has been shown to decrease one's immune system. Therefore, unless clients are getting enough rest/recovery, eating healthy foods, and taking care of their body, they are likely at a greater risk for getting sick than those who do not train for endurance sports (295).

When an individual gets sick from a cold or flu, the body undergoes a physiological response to protect it and help it heal. Unfortunately, the mechanisms that help to heal the body also decrease its ability to perform at peak level. Below are several physiological responses that occur with illness (296):

1. The body breaks down muscle (catabolism)
    - Studies done on individuals with fevers showed microscopic damage to muscles.

- Muscle strength decreases, and it can take up to two weeks to regain strength (from a three-day illness associated with a fever).

2. Fat metabolism is decreased
    - Body is primarily utilizing muscle protein for fuel, not fat.
3. Aerobic metabolism is diminished
    - VO_2 max and lactate threshold will decrease.

When fighting off a cold or something more serious like the flu, exercise should not be done. The focus must be on rest and recovery. Even if a client were to exercise when sick, there would be little to no improvement in fitness, and likely the opposite would occur.

Rest is often hard for clients to adhere to as they view lost training days as exactly that – lost. It is your responsibility to inform them that they will not see any physiological gains when training while sick and that by doing so they will only extend their time being "out of commission."

Guidelines

- If clients are symptomatic (e.g., runny nose, elevated temperature, congestion, body ache, chills, etc.) they should not exercise.

    - If a client thinks he or she might have a simple cold but is asymptomatic, it is best to either not exercise, or exercise at an extremely low level for a short duration.

- If a head cold has "moved" into the chest, it is imperative not to exercise.

- Rest and recovery is the best bet until the client feels better and shows no signs of illness.

HEAT-RELATED ILLNESSES

Racing or training in hot environments can lead to heat-related illnesses. There are three categories of heat-related illnesses:

1. Heat Cramps
2. Heat Exhaustion

3. Heat Stroke

The body has approximately 2 million to 4 million sweat glands (595). Sweat secreted by the glands is cooled via evaporation. The purpose of cooling the skin is to cool the blood that has been shunted to the skin from deeper aspects of the body (organs, muscles) (595). However, if exercising in humid conditions, the degree of evaporation is substantially lessened. Thus, exercising in hot and humid conditions can prove dangerous because of a lack of evaporative cooling.

> Wiping away sweat before it has had a chance to evaporate greatly reduces the cooling benefit of sweating.

Heat Cramps

Characterized by muscle cramps and sweating, heat cramps (not exercise-based) are often treated with cessation of activity, moving to a cool, shaded area, and drinking an electrolyte-based beverage. Stretching and massaging the cramped area can also help to alleviate acute pain.

Heat Exhaustion

This occurs when the body's natural temperature regulation system begins to break down. Because of a lack of hydration and minerals, the capillaries of the body reduce in size, thereby reducing the body's effectiveness to cool itself. Common symptoms include (297):

- Dizziness
- Heavy sweating
- Confusion
- Nausea
- Weak, rapid pulse
- Low blood pressure
- Excessive thirst
- Hyperventilation
- Loss of appetite
- Anxiety

The client must stop activity and move to a shady, cool location. The individual should be given an electrolyte drink and cooled with a fan and/or wet cloths. Additionally, the person should be positioned in the supine position with legs elevated. Medical personnel should be called to check and monitor the client.

Heat Stroke

This is the most serious of the three categories of heat sickness and can be fatal. While an individual may have experienced heat cramps and exhaustion prior to heat stroke, it is not always the case. Heat stroke is caused by the body's lack of water and electrolytes. Effectively, the body is shutting down. Symptoms include (297).

- Rapid heart rate
- No sweating
- High body temperature (>103)
- Red, dry skin
- Confusion
- Vomiting
- Nausea
- Erratic behavior
- Difficulty breathing
- Constricted pupils

The most important thing to do is lower the individual's body temperature immediately and get emergency personnel on-site. Call 911.

Prevention

By staying properly hydrated, acclimating to heat gradually, taking breaks, wearing proper clothing (e.g., light colors, wicking), and being aware of the body's response to the heat, the chance of developing a serious heat-related injury can be greatly mitigated. Prehydrating with cold fluids prior to training or racing in the heat has been shown to reduce one's core temperature (592).

Heat Acclimatization

Just as preparing to race at high altitude requires acclimatization, so does preparing to race or train in excessive heat. When training a client for a race that is most likely going to be undertaken in much warmer temperatures than is being trained in, proper heat acclimatization is critical. A few important factors to consider:

1. Individuals will differ greatly in their ability to deal with heat and humidity.
2. The more body fat an individual has, the less efficient the person will be at dissipating body heat.
3. Generally, the larger the runner, the more energy is required to reduce body heat. This is because the smaller the individual, the larger the skin surface compared with body volume. This enables small individuals to dissipate heat better than large individuals (487).
    a. Paula Radcliffe, who at five-foot-eight was a very tall elite marathoner, performed well only in marathons that were in cool temperatures. This is because she was less efficient at dissipating heat compared with her shorter competitors (488).
4. Wind speed, humidity, and shade all affect the runner (311).
    a. Sweat must evaporate to cool the body. Therefore in humid conditions where sweat does not evaporate as well as in nonhumid conditions, the chance for overheating increases.
5. Any heat acclimatization must be done slowly, as it is a gradual process.
    a. Start with 10–15 minutes/day of light exercise and progressively increase the duration and intensity of exercise sessions (595).
6. Heat acclimation takes approximately two to four weeks (90).
    a. After 10 days of heat acclimatization, the capacity for sweating nearly doubles (596).
7. Novice endurance athletes and deconditioned individuals are often the most negatively affected by hot weather.
8. The physiological benefits of heat acclimatization are lost between two and three weeks after returning to a more temperate environment (594).

The progression of heat acclimation should be steady and gradual and should be done sparingly, if at all, during the taper period. Simulated heat can be produced either by increasing the temperature (training indoors with thermostat turned up) and/or wearing additional clothing. Generally it is better to increase the temperature versus wear more clothing, as the body will be able to cool itself better because of evaporation. The progression is a function of time and temperature. It is important that when performing heat acclimatization training, an increase in hydration is done simultaneously. During heat acclimatization, one's heart rate and sweat rate needs to be carefully monitored, as the heat will increase one's heart rate. If your client is going to undertake heat acclimatization training, it is advised for them to speak with a sports medicine physician about the best method to elicit the best results safely.

Individuals who are heat acclimatized can sweat up to 4 liters/hour, while those who are not acclimatized will sweat approximately 1.5 liters/hour (732). This demonstrates the increased efficiency regarding thermoregulation due to heat acclimatization training.

HYPOTHERMIA

Hypothermia is the opposite of heat-related illnesses. Hypothermia is the body's inability to maintain normal metabolic functioning and is representative of a body temperature of 95 degrees F or less (normal body temperature = 98.6 degrees F).

If a client is training or racing in cold temperatures without proper clothing, hypothermia can also set in.

Physiologically, the body looks to preserve heat in the most vital areas. Therefore, the body moves blood away from the extremities to more vital, life-sustaining organs. Heat production can be increased two to four times through muscle contractions (299). This is why the body shivers – to produce heat.

Water increases heat loss. Therefore, wearing wicking clothing is very important when running in cold weather. When clothing becomes wet with sweat, body heat loss is accelerated.

Like heat-related illnesses, there are several stages of hypothermia (451):

Mild Hypothermia

The most common visible symptom is shivering and cold or numb extremities due to the constriction of blood vessels (vasoconstriction). Moving to a warm area, taking off wet clothes, and putting on dry and adequate amounts of clothing are the most common solutions to eliminate mild hypothermia.

Moderate Hypothermia

The most common visible symptoms are excessive shivering, a loss of muscular coordination, and extremities – including the lips and ears – turning blue because of a lack of blood flow to these areas. An individual must be immediately moved to a warm area and changed into dry clothing. A warm hat is also suggested. Medical attention is required.

Severe Hypothermia

At this point, the body begins to shut down and is represented by the heart rate, blood pressure, and respiration rate all declining substantially. An individual will also likely have a hard time walking and talking, and their cognitive ability is typically compromised. Heart and respiration rates continue to fall in relation to moderate hypothermia (300). Unless an individual is moved to a warm area and medical attention is provided, the organs of the body will shut down and death will result. Medical attention must be provided.

Prevention

First and foremost, clients must be aware of their environment as well as the weather forecast for the area they are training and racing in. Proper clothing selection is critical to avoiding hypothermia. They should wear sweat-wicking fabrics and dress in layers. Additionally, extremities and normally exposed body parts (e.g., ears, nose) must be covered in extremely cold conditions. When training in cold conditions, clients must always have a partner to train with in case of an emergency. It is a good idea to make sure they are training in an area near civilization in case of an emergency, and they must always tell someone of their planned route. If at all possible, it is advised to train in a concentrated area near their residence instead of covering a large geographic area. If clients get too cold, they must know that they are not far from home. It is also a good idea to carry a cell phone in case of an emergency.

Regardless of the planned training for the day, clients must be attentive to their body's response to the cold, and if they feel they are getting too cold, they must head for home or seek shelter in a warm environment.

HIGH ALTITUDE

If a running race is at high altitude (6,000/ft or above) and your client lives at or near sea level, it would be advantageous to arrive a minimum of three or four days ahead of the race to acclimate. Individuals acclimate at different rates. Because of the decreased oxygen level at altitude, the body cannot work as hard. Many people who go from sea level to high altitude succumb to altitude sickness. This is especially the case for those who put forth high levels of physical exertion. Common symptoms of altitude sickness include:

- Headaches
- Dizzy and light-headed
- Nausea
- Excessive shortness of breath
- Weakness and fatigue
- Bloody nose (often due to dry air)

Another way to prepare for an event at altitude is to sleep in an altitude tent prior to the event. To recap what was discussed in the hypoxic training section, an altitude tent allows a user to simulate sleeping at high altitude, thereby acclimating prior to leaving for an event. Altitude tents can sometimes be rented. Clients must ensure that use of an altitude tent is legal based on their sport's national governing body.

If clients are traveling by car to an event, staying overnight at an elevation between their origin and the event's elevation can help in the acclimatization process.

Individuals tend to become dehydrated faster at high altitudes. Therefore, consuming slightly higher amounts of water might help in reducing high-altitude sickness (298). Generally speaking, the longer one is at altitude, the more acclimated the person becomes. However, the only reliable way to reduce altitude sickness is to go to a lower altitude.

Hydration-related illnesses/conditions such as *dehydration* and *hyponatremia* are discussed in greater detail in the *Sports Nutrition* module.

❖ SUMMARY

INJURY

- In emergency situations, emergency personnel must be called (911).
- As a certified UESCA coach, you must be CPR/AED certified.
- To mitigate the chance for injury, be attentive to the following areas:
    - Volume increases
    - Intensity-level increases
    - Monitoring fatigue level
    - Integrating cross-training
- Running has a high injury rate, primarily because of the impact nature of the activity.
- Treadmills have the lowest impact on the feet, followed by grass surfaces. The rest of the surfaces (i.e., cement, asphalt, dirt) are comparable in impact forces.
- There is no definitive cause of side stitches.
- There are many potential origins of shin splints. If your client has developed a shin splint, aside from not running, the person should seek out a sports physician or physical therapist.
- If your client suffers a sports-related injury, the individual should seek out a specialist such as a physical therapist or a sports medicine physician.
- As a coach, when dealing with injury, you must always work within your scope of knowledge and practice.
- You cannot diagnose a client's injury.
    - The only treatment you can advise to a client is **R**est–**I**ce–**C**ompression–**E**levation (RICE). Beyond RICE, recommending treatments as seemingly benign as heat need to be avoided.

ILLNESS

- If your client has a cold and is symptomatic, the person should not exercise.
- When sick, the body becomes catabolic and fat metabolism decreases.
- The three types of heat illnesses are: heat cramps, heat exhaustion, and heat stroke. Heat stroke is the most serious and can be fatal.
- Heat acclimatization must be done gradually and safely and should be advised by a physician.

- The three levels of hypothermia are mild, moderate, and severe. Like heat stroke, severe hypothermia can be fatal.
- When exercising in extreme heat or cold, individuals must be very aware of their body's condition.
- To avoid altitude sickness, a progressive acclimatization program is strongly suggested.
- Individuals acclimate at different rates.
- As a coach, when dealing with illness, you must always work within your scope of knowledge and practice.